

→ Solos limites laterais. Corrigimos tirar as conclusões re. tende 0 + ou - infinito.

1) a) $\lim_{x \rightarrow 1^-} \frac{1}{x-1} = -\infty$

$x = 0,9 \rightarrow \frac{1}{0,9-1} = \frac{1}{-0,1} = -10$
 $x = 0,99 \rightarrow \frac{1}{0,99-1} = \frac{1}{-0,01} = -100$

b) $\lim_{x \rightarrow 1^+} \frac{1}{x-1} = +\infty$

$x = 1,1 \rightarrow \frac{1}{0,1} = 10$
 $x = 1,01 \rightarrow \frac{1}{0,01} = 100$

c) Librenamos que as laterais desse limite estão divergindo, ou seja, esse limite não existe.

* Chegamos numa conclusão, pois as laterais estão divergindo para o infinito.

d) $\lim_{x \rightarrow 1^-} \frac{2x+1}{x-1} = -\infty$

$x = 0,1 \rightarrow \frac{1,2}{-0,9} = -\frac{12}{9}$
 $x = 0,01 \rightarrow \frac{1,02}{-0,99} = -\frac{102}{99}$

e) $\lim_{x \rightarrow 1^+} \frac{2x+1}{x-1} = +\infty$

$x = 1,1 \rightarrow \frac{3,2}{0,1} = 32$
 $x = 1,01 \rightarrow \frac{3,02}{0,01} = 302$

f) $\lim_{x \rightarrow 1} \frac{3x+2}{(x-1)^2} = +\infty$

$x = 1,1 \rightarrow \frac{5,3}{(0,1)^2} = \frac{5,3}{0,01} = \frac{53 \cdot 10^{-1}}{10^{-2}} = 53 \cdot 10 = 530$
 $x = 0,9 \rightarrow \frac{3,8}{10^{-2}} = \frac{38 \cdot 10^{-1}}{10^{-2}} = 38 \cdot 10^{-1-(-2)} = 38 \cdot 10 = 380$

g) $\lim_{x \rightarrow 2} \frac{1-x}{(x-2)^2} = -\infty$

$x = 2,1 \rightarrow -110$
 $x = 1,9 \rightarrow -90$

h) $\lim_{x \rightarrow 0} \frac{3x+2}{(x-1)^2} = \infty$

i) $\lim_{x \rightarrow 0} \frac{3x^2-5x+2}{x^2} = +\infty$

$x = 0,1 \rightarrow 153$
 $x = -0,1 \rightarrow 553$

l) $\lim_{x \rightarrow 1^+} \frac{\sqrt{x^2-1}}{x-1} = +\infty$

* Podemos observar que sempre irá tender ao infinito, ainda mais pelo limite tender o 1+

j) $\lim_{x \rightarrow -2} \frac{2x^2+5x-3}{|x+2|} =$

$|x+2| = \begin{cases} x+2, & x \geq -2 \\ -(x+2), & x < -2 \end{cases}$

k) $\lim_{x \rightarrow -1} \frac{5x+3}{|x+1|} = -\infty$

$|x+1| = \begin{cases} x+1, & x \geq -1 \\ -(x+1), & x < -1 \end{cases}$

$x^+ = -2,7$ * módulo negativo

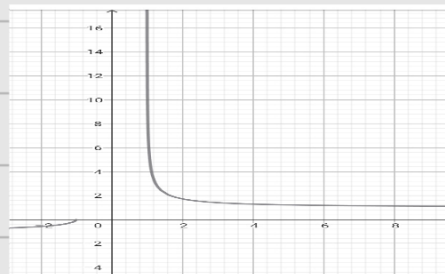
$\frac{2x^2+5x-3}{-(x+2)} = \frac{0,81-10,5-3}{0,1} = -46,9$

$x^+ = -1,1 \rightarrow \frac{5x+3}{-(x+1)} = \frac{-5,5+3}{-0,1} = -25$

$x^- = -1,9$ * módulo positivo

$\frac{2x^2+5x-3}{x+2} = \frac{7,22-9,5-3}{-0,1} = -53,8$

$x^- = -0,9 \rightarrow \frac{5x+3}{x+1} = \frac{-4,5+3}{0,1} = -15$



2) a) $\lim_{x \rightarrow \infty} (4-5x) = -\infty$

b) $\lim_{x \rightarrow -\infty} (4-x^2) = -\infty$

c) $\lim_{x \rightarrow -\infty} (8-x^3) = +\infty$

d) $\lim_{x \rightarrow \infty} (3x^3-4) = +\infty$

e) $\lim_{x \rightarrow -\infty} \sqrt{x^2-3x+5} = \infty$

$\sqrt{(x^4)(1-\frac{3}{x}+\frac{5}{x^2})} = |x| \cdot 1 = +\infty$

→ A raiz de algo positivo é sempre positiva
 → $\sqrt{(x^2)} = |x|$

f) $\lim_{x \rightarrow \infty} \frac{3x+2}{5x-1} \rightarrow \frac{\infty}{\infty} = ?$

$\therefore \lim_{x \rightarrow \infty} \frac{3}{5} //$

g) $\lim_{x \rightarrow \infty} \frac{5-4x}{2x-3} \rightarrow \frac{-\infty}{\infty} = ?$

h) $\lim_{x \rightarrow \infty} \frac{5x^2-4x+3}{3x+2} = \frac{\infty}{\infty} = ?$

$\frac{3x+2}{5x-1} = \frac{x \cdot (3+\frac{2}{x})}{x \cdot (5-\frac{1}{x})} = \frac{3+\frac{2}{x}}{5-\frac{1}{x}} = \frac{3}{5}$

$\lim_{x \rightarrow \infty} \frac{5-4x}{2x-3} = \frac{x \cdot (\frac{5}{x}-4)}{x \cdot (2-\frac{3}{x})} = \frac{-4}{2} = -2$

$\lim_{x \rightarrow \infty} \frac{5x^2-4x+3}{3x+2} = \frac{x \cdot (5x-4+\frac{3}{x})}{x \cdot (3+\frac{2}{x})} =$

$= \lim_{x \rightarrow \infty} \frac{5x-4}{3} = \infty //$

i) $\lim_{x \rightarrow \infty} \frac{4x-1}{3x^2+5x-2} = \frac{\infty}{\infty} = ?$

j) $\lim_{x \rightarrow -\infty} \frac{x^2-1}{x^3+1} = \frac{\infty}{-\infty} = ?$

* Aqui se divide +
 zero - ou 0.
 Eu de 0, pois estamos
 trabalhando com -oo no
 denominador.

$\lim_{x \rightarrow \infty} \frac{4x-1}{3x^2+5x-2} = \frac{x \cdot (4-\frac{1}{x})}{x \cdot (3x+\frac{5}{x}-\frac{2}{x^2})} = \frac{4-\frac{1}{x}}{3x+\frac{5}{x}-\frac{2}{x^2}} = \frac{4}{3x} = 0$

$\lim_{x \rightarrow -\infty} \frac{x^2-1}{x^3+1} = \frac{x^2 \cdot (1-\frac{1}{x^2})}{x^3 \cdot (1+\frac{1}{x^3})} = \frac{1-\frac{1}{x^2}}{x \cdot (1+\frac{1}{x^3})} = \frac{1}{x} = 0$

3) a) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2-2x+2}}{x+1} = \frac{\infty}{\infty} = ?$

b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2-2x+2}}{x+1} = \frac{\sqrt{x^2} \cdot \sqrt{1-\frac{2}{x}+\frac{2}{x^2}}}{x+1} =$

c) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+x+1}}{x+1} = \frac{\sqrt{x^2} \cdot \sqrt{1+\frac{1}{x}+\frac{1}{x^2}}}{x \cdot (1+\frac{1}{x})} =$

$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2-2x+2}}{x+1} = \frac{\sqrt{x^2} \cdot \sqrt{1-\frac{2}{x}+\frac{2}{x^2}}}{x+1} =$

$= \frac{-x \cdot \sqrt{1-\frac{2}{x}+\frac{2}{x^2}}}{x \cdot (1+\frac{1}{x})} = \frac{-1}{1} = \lim_{x \rightarrow -\infty} -1$

$= \frac{x \cdot \sqrt{1+\frac{1}{x}+\frac{1}{x^2}}}{x \cdot (1+\frac{1}{x})} = \frac{1}{1} = \lim_{x \rightarrow \infty} 1$

$= \frac{x \cdot \sqrt{1-\frac{2}{x}+\frac{2}{x^2}}}{x \cdot (1+\frac{1}{x})} = \frac{1}{1} = \lim_{x \rightarrow \infty} 1$

* O detalhe do módulo numa
 questão é o sinal.

d) $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 - 2x + 2}}{x + 1} =$
 $= \frac{-x \cdot \sqrt{1 - \frac{2}{x} + \frac{2}{x^2}}}{x \cdot (1 + \frac{1}{x})} = \frac{-1}{1} = -1$

e) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x - 6}{\sqrt{x^4 + 1}} = \frac{2x^2 - 3x - 6}{\sqrt{x^4} \cdot \sqrt{1 + \frac{1}{x^4}}} =$
 $= \frac{x^2 \cdot (2 - \frac{3}{x} - \frac{6}{x^2})}{x^2 \cdot (1 + \frac{1}{x^4})} = \lim_{x \rightarrow \infty} \frac{2}{1} = 2$

f) $\lim_{x \rightarrow -\infty} \frac{2x^2 - 3x - 6}{\sqrt{x^4 + 1}} = \frac{2x^2 - 3x - 6}{\sqrt{x^4} \cdot \sqrt{1 + \frac{1}{x^4}}} =$
 $= \frac{x^2 \cdot (2 - \frac{3}{x} - \frac{6}{x^2})}{x^2 \cdot (1 + \frac{1}{x^4})} = \lim_{x \rightarrow -\infty} \frac{2}{1} = 2$

g) $\lim_{x \rightarrow \infty} \frac{x^2}{1 + x \sqrt{x}} = \frac{x^2}{x \cdot (\frac{1}{x} + \sqrt{x})} =$
 $= \frac{x}{\frac{1}{x} + \sqrt{x}} = \frac{x}{\sqrt{x}} = \sqrt{x} = \lim_{x \rightarrow \infty} \infty$

h) $\lim_{x \rightarrow -\infty} \frac{x + \sqrt{x}}{x^2 + 1} = \frac{\infty - \infty}{\infty + 1} = 0$
 $\lim_{x \rightarrow -\infty} \frac{x + \sqrt{x}}{x^2 + 1} = \frac{x \cdot (1 + \frac{1}{\sqrt{x}})}{x^2 \cdot (1 + \frac{1}{x^2})} = \frac{1}{x} = \frac{1}{-\infty} = 0 \Rightarrow \lim_{x \rightarrow -\infty} 0 //$

4) a) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x + 2} - x) = \infty - \infty?$

$\lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 3x + 2} - x) \cdot (\sqrt{x^2 + 3x + 2} + x)}{\sqrt{x^2 + 3x + 2} + x} =$
 $= \frac{x^2 + 3x + 2 - x^2}{\sqrt{x^2 + 3x + 2} + x} = \frac{x \cdot (3 + \frac{2}{x})}{x \cdot (\sqrt{1 + \frac{3}{x} + \frac{2}{x^2}} + 1)} = \frac{3}{2}$

b) $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 3x + 2} - x) =$

$\lim_{x \rightarrow -\infty} \frac{(\sqrt{x^2 + 3x + 2} - x) \cdot (\sqrt{x^2 + 3x + 2} + x)}{\sqrt{x^2 + 3x + 2} + x} =$
 $= \lim_{x \rightarrow -\infty} \frac{x^2 + 3x + 2 - x^2}{\sqrt{x^2 + 3x + 2} + x} = \lim_{x \rightarrow -\infty} \frac{3}{-\infty} = 0 //$

c) $\lim_{x \rightarrow \infty} (\sqrt{x+4} - \sqrt{x-2}) = \infty - \infty?$

$\lim_{x \rightarrow \infty} \frac{(\sqrt{x+4} - \sqrt{x-2}) \cdot (\sqrt{x+4} + \sqrt{x-2})}{\sqrt{x+4} + \sqrt{x-2}} = \frac{(x+4) - (x-2)}{\sqrt{x+4} + \sqrt{x-2}} =$
 $= \lim_{x \rightarrow \infty} \frac{6}{\sqrt{x+4} + \sqrt{x-2}} = \frac{6}{\infty} = \lim_{x \rightarrow \infty} 0 //$

d) $\lim_{x \rightarrow \infty} |\sqrt{x+1} - \sqrt{x-1}| = \infty - \infty?$

$\lim_{x \rightarrow \infty} \frac{(\sqrt{x+1} - \sqrt{x-1}) \cdot (\sqrt{x+1} + \sqrt{x-1})}{\sqrt{x+1} + \sqrt{x-1}} =$
 $= \frac{(x+1) - (x-1)}{\sqrt{x+1} + \sqrt{x-1}} = \frac{2}{\sqrt{x+1} + \sqrt{x-1}} = \frac{2}{\infty} = 0 //$

e) $\lim_{x \rightarrow \infty} \sqrt{x^2 - 4x + 5} - \sqrt{x^2 - 3x + 4} = \infty - \infty?$

$\lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 - 4x + 5} - \sqrt{x^2 - 3x + 4}) \cdot (\sqrt{x^2 - 4x + 5} + \sqrt{x^2 - 3x + 4})}{\sqrt{x^2 - 4x + 5} + \sqrt{x^2 - 3x + 4}} =$
 $= \frac{(x^2 - 4x + 5) - (x^2 - 3x + 4)}{\sqrt{x^2 - 4x + 5} + \sqrt{x^2 - 3x + 4}} = \frac{-x + 1}{\sqrt{x^2 - 4x + 5} + \sqrt{x^2 - 3x + 4}} = \frac{-x + 1}{x + x} = \frac{-x + 1}{2x} = \frac{-1 + \frac{1}{x}}{2} = -\frac{1}{2} //$

f) $\lim_{x \rightarrow \infty} x \cdot \sqrt{x^2 + 4} = \infty - \infty?$

$\lim_{x \rightarrow \infty} \frac{x \cdot \sqrt{x^2 + 4}}{1} = \frac{x \cdot \sqrt{x^2 + 4}}{x + \sqrt{x^2 + 4}} =$
 $= \frac{x^2 - (x^2 + 4)}{x + \sqrt{x^2 + 4}} = \frac{-4}{x + \sqrt{x^2 + 4}} = \frac{-4}{\infty} = 0 //$

g) $\lim_{x \rightarrow -\infty} \frac{x + \sqrt{x^2 + 4}}{1} = \frac{x + \sqrt{x^2 + 4}}{x - \sqrt{x^2 + 4}} = \frac{x^2 - (x^2 + 4)}{x - \sqrt{x^2 + 4}} =$
 $= \frac{-4}{x - \sqrt{x^2 + 4}} = \frac{-4}{x - x} = \frac{-4}{0} = \infty //$

5) a) $f(x) =$

$Df(x) = \mathbb{R} - \{3\}$

$\lim_{x \rightarrow 3^+} \frac{2x+1}{x-3} = +\infty$

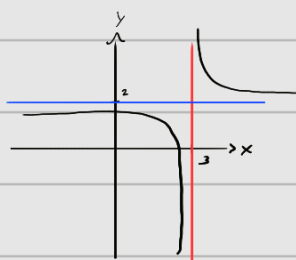
$\lim_{x \rightarrow 3^-} \frac{2x+1}{x-3} = -\infty$

$x=3$ é uma assíntota vertical

$\lim_{x \rightarrow \infty} \frac{2x+1}{x-3} = \frac{x \cdot (2 + \frac{1}{x})}{x \cdot (1 - \frac{3}{x})} = 2 //$

$\lim_{x \rightarrow -\infty} \frac{2x+1}{x-3} = \frac{x \cdot (2 + \frac{1}{x})}{x \cdot (1 - \frac{3}{x})} = 2 //$

$y=2$ também é uma assíntota horizontal.



b) $f(x) = 1 + \frac{1}{x^2}$

Verticais:

$D = \mathbb{R} - \{0\}$ * Vamos testar os valores próximos de 0.

$\lim_{x \rightarrow 0^-} 1 + \frac{1}{x^2} = +\infty$

$x = -0.1 \rightarrow 101$

$x = -0.01 \rightarrow 10^4 + 1$

$\lim_{x \rightarrow 0^+} 1 + \frac{1}{x^2} = +\infty$

$x = 0.1 \rightarrow 101$

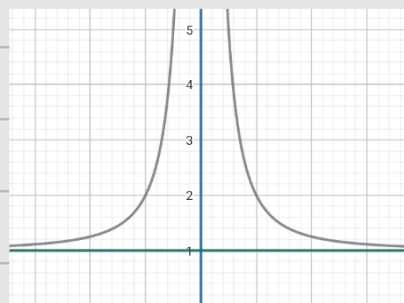
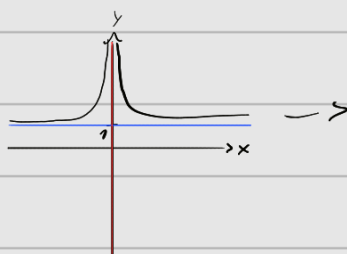
$x = 0.01 \rightarrow 10^4 + 1$

* Portanto, $x=0$ é uma assíntota vertical

horizontais:

$\lim_{x \rightarrow \infty} 1 + \frac{1}{x^2} = 1$

* Portanto, $y=1$ é uma assíntota horizontal



$$c/d) f(x) = \frac{-3x}{\sqrt{x^2+3}}$$

Verticais:

$$D = \mathbb{R} \text{ * } \emptyset \text{ denominador não zero com } \sqrt{-3}$$

ou não, não temos assíntotas Verticais

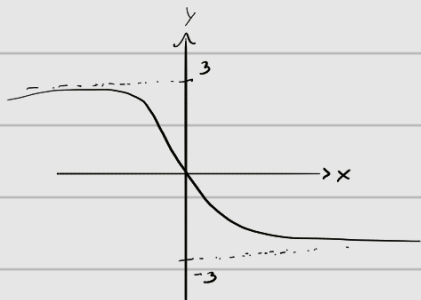
horizontais:

$$\lim_{x \rightarrow \infty} \frac{-3x}{\sqrt{x^2+3}} = \frac{-3x}{\sqrt{x^2} \sqrt{1+\frac{3}{x^2}}} = \frac{-3x}{x} = -3$$

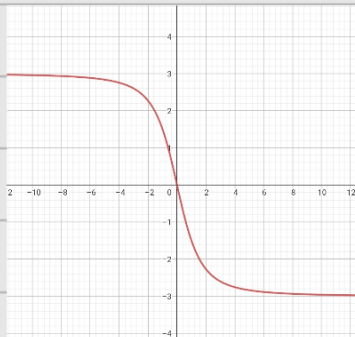
$$\lim_{x \rightarrow -\infty} \frac{-3x}{\sqrt{x^2+3}} = \frac{-3x}{-x} = 3$$

* Temos duas assíntotas Horizontais

$$y = 3 \text{ e } y = -3$$



→



$$e) f(x) = \frac{x}{\sqrt{x^2-9}} = \frac{x}{\sqrt{x^2} \sqrt{1-\frac{9}{x^2}}}$$

Verticais:

$$D = \mathbb{R} - \{\pm 3\}$$

* Em 3 e -3 temos as assíntotas verticais, mas não se entende nem

comportamento, para resolver a gráfica.

$$\lim_{x \rightarrow 3^-} \frac{x}{\sqrt{x^2-9}} = \frac{3}{0^+} = \infty$$

* Note que foge os valores convergindo

$$\lim_{x \rightarrow 3^+} \frac{x}{\sqrt{x^2-9}} = \frac{3}{0^+} = -\infty$$

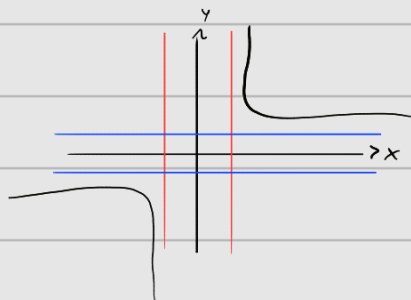
Não é zero, mas um valor que se aproxima muito.

Horizontais:

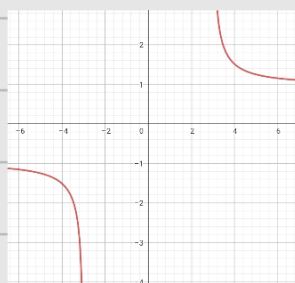
$$\lim_{x \rightarrow 3} \frac{x}{\sqrt{x^2} \sqrt{1-\frac{9}{x^2}}} = \frac{x}{x \cdot 1} = 1$$

$$\lim_{x \rightarrow -3} \frac{x}{\sqrt{x^2} \sqrt{1-\frac{9}{x^2}}} = \frac{x}{-x \cdot 1} = -1$$

* Portanto, verificamos $x=3$, $x=-3$, $y=1$ e $y=-1$



→



$$f) 3xy - 2x - 4y - 3 = 0 \rightarrow y = \frac{2x+3}{3x-4}$$

Verticais:

$$D = \mathbb{R} - \{\frac{4}{3}\}$$

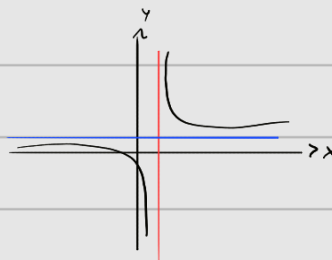
$$\lim_{x \rightarrow \frac{4}{3}^-} \frac{2x+3}{3x-4} = \frac{5}{-1} = -\infty$$

$$\lim_{x \rightarrow \frac{4}{3}^+} \frac{2x+3}{3x-4} = \frac{5}{1} = +\infty$$

horizontal:

$$\lim_{x \rightarrow \infty} \frac{2x+3}{3x-4} = \frac{x(2+\frac{3}{x})}{x(3-\frac{4}{x})} = \frac{2}{3}$$

$$\lim_{x \rightarrow -\infty} \frac{2x+3}{3x-4} = \frac{2}{3}$$



$$g) x^4 - 3y^4 - 9x = 0 \rightarrow y = \frac{\sqrt[4]{3\sqrt{x}}}{\sqrt{x-3}}$$

Verticais:

$$D = \mathbb{R} - \{3\}$$

$$\lim_{x \rightarrow 3^+} \frac{\sqrt[4]{3\sqrt{x}}}{\sqrt{x-3}} = +\infty$$

Horizontais:

$$\lim_{x \rightarrow \infty} \frac{\sqrt[4]{3\sqrt{x}}}{\sqrt{x-3}} = 3 \sqrt{\frac{x}{(x-3)}} =$$

$$3 \sqrt{\frac{x}{x(1-\frac{3}{x})}} = 3$$

3 é assíntota vertical.

→

